

Risk-Stratified Explainability of Attorney Involvement in Workers' Compensation Claims

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Abstract—Workers' compensation claims can often result in legal disputes, where injured workers seek attorney representation to secure full benefits. Although useful for navigating legal complexities, attorney involvement is associated with prolonged claim durations and higher costs with workers' compensation claims. Prior studies generally model legal representation as a binary outcome, limiting insights into the probability distribution of litigation across the cases. Using a private dataset of workers' compensation claims, this study conceptualizes attorney involvement as a machine learning-based risk stratification approach. First, we estimate claim-level risk using a CatBoost classifier and evaluate model performance with both discrimination and calibration metrics, including ROC-AUC, PR-AUC, Brier score, and Expected Calibration Error. Then, to examine litigation patterns, we use the predicted probability scores to segment the claims into three rank-based groups: low, medium, and high risk. Lastly, we measure the top feature contributions within each group using SHAP (Shapley Additive Explanations) values to assess how key predictors change across the risk levels. Our findings indicate that injury descriptors, jurisdiction, and age demographics are consistently ranked among the most influential features, but their effects in magnitude and direction vary as risk escalates. Overall, the study demonstrates the value of probabilistic modeling for identifying drivers of litigation risk in workers' compensation claims, which can support more targeted case management.

Index Terms—Occupational injury, Workers' Compensation, Shapley Additive Explanations, Machine learning

I. INTRODUCTION

Employees with a job-related injury must file a workers' compensation claim to receive benefits, such as wage replacement and medical care coverage. However, legal disputes may arise from complex cases and can involve insurance denials, unsatisfactory settlements, and medical complications. These situations can place significant financial and administrative strain on workers, employers, and insurers. To overcome these challenges, the injured employee may hire attorney representation to manage legal proceedings and earn a higher payout [1]. Conversely, attorney involvement in workers' compensation claims is associated with longer claim durations and increased costs [2]. Therefore, identifying claims with an elevated likelihood of attorney involvement at an early stage is important for improving resource allocation and enabling proactive case management.

Previous studies largely rely on traditional statistical methods to investigate predictors of attorney representation and treat the outcome as a binary event without analyzing differences in litigation risk across the cases [2], [3]. However,

advanced machine learning models can improve predictive performance, handle high-dimensional claim data, and capture nonlinear relationships. Rather than framing the problem as a binary classification task, a risk modeling approach can be used by training a machine learning model to produce probability estimates and ordering claims by risk. This ranking can provide an enhanced view of features that influence attorney likelihood across the risk distribution.

In this study, we model attorney involvement using a CatBoost classifier trained on 48,130 workers' compensation claims from a private dataset. Instead of using binary outcomes, the model generates probability estimates that are interpreted as risk scores. Based on these scores, the test set is ranked and segmented into three risk regimes: the bottom 20% as low risk, the middle 60% as medium risk, and the top 20% as high risk. To interpret model behavior, we examine feature importance using mean absolute and signed Shapley Additive Explanations (SHAP) values to determine the drivers across the risk spectrum. This SHAP analysis highlights differences in both the magnitude and direction of feature contributions across low-, medium-, and high-risk claims. Our findings show that several predictors display notable changes in directional influence between the low- and high-risk groups, indicating that their impact depends on the specific level of risk. To the best of our knowledge, this is the first study to explore risk stratification and SHAP contributions for predicting attorney involvement in workers' compensation claims.

The remainder of this paper is organized as follows. Section II reviews related works. Section III describes the dataset, CatBoost modeling, and SHAP analysis. Section IV examines the risk-stratified feature importance results. Section V discusses the implications for attorney involvement. Finally, Section VI concludes with the findings and future work.

II. RELATED WORKS

In the United States, workers' compensation is a state-mandated insurance system with jurisdiction-specific rules, incentives, and administrative structures. This program structure complicates the creation of a unified national database on workers' compensation claims and contributes to substantial variation in litigation rates across states. These regulatory differences influence both benefit outcomes and the likelihood that claims become disputed or legally represented [4].

Past studies highlight the relationship between attorney representation and several claim characteristics using statistical

modeling. For example, Bernacki and Tao compare 6,929 claims from Louisiana using descriptive analysis and logistic regression. The authors find that attorney involvement is associated with longer claim durations and higher costs after controlling for workdays lost due to injury [2]. Borba and Appel employ an econometric model for California workers with permanent disabilities and conclude that attorney hiring is influenced by injury seriousness, worker and job characteristics, and perceived quality of claim handling [5]. With Michigan-based data, Roberts similarly uses logistic regression to determine that the number of weeks of temporary total benefit payments and the number of days between the injury date and first benefit payment are decisive factors for predicting claim disputes [3]. Together, these findings suggest that legal representation is not solely decided by injury severity and instead arises from a combination of clinical and administrative variables. However, these works primarily rely on traditional approaches to capture average effects and provide limited insight into how escalation dynamics vary across the risk distribution.

Although research focused on workers' compensation litigation propensity remains limited, machine learning methods have become more widely applied to related outcomes in the occupational injury domain. For instance, Davoudi Kakhki et al. predict cost severity classification with gradient boosting trees, support vector machines, and Naïve Bayes [6]. Their work includes similar features to our dataset, namely injured body part, nature of injury, cause of injury, and age of the injured worker. Mathews investigates whether support vector machines, regularized logistic regression, Naïve Bayes, and CART (Classification and Regression Trees) are well-suited for cost severity prediction [7]. Along with logistic regression, Lee et al. implement four advanced machine learning models XGBoost, CatBoost, LightGBM, and Random Forest to predict post-injury dissatisfaction [8]. Furthermore, the authors utilize a SHAP analysis to identify key predictors and improve model interpretability. Collectively, these studies demonstrate that administrative claim data can effectively support accurate predictive modeling of outcomes related to occupational injuries.

Existing literature establishes attorney involvement as an important escalation outcome and shows that the potential for predictive modeling with workers' compensation claims data [2], [5]. However, treating attorney involvement as a binary endpoint limits insight into risk progression. Even when machine learning is applied, these studies often prioritize classification performance over risk modeling. Furthermore, model interpretability and feature importance indicators are commonly viewed as global summaries, obscuring variation in predictor effects across risk levels. These gaps inform the design of our approach. Instead of framing attorney involvement as a binary outcome, we model our target as ordered probability scores and organize claims into low-, medium-, and high-risk regimes. Lastly, we examine feature contributions across the three regimes, evaluating the absolute magnitude and directional effect of each feature.

III. METHODOLOGY

A. Data Preprocessing

This study uses the Structured and Textual Records for Injury Data Exploration (STRIDE) database, an anonymized repository of workers' compensation claims, which contains a structured tabular dataset of claim-level attributes and a textual dataset of adjuster notes [9]. Our study focuses solely on the tabular dataset in order to isolate litigation risk dynamics using standardized claim variables. The tabular data is made up of 230,833 workers' compensation claims from January 2002 to December 2020 across 32 states and includes injury descriptors, worker demographics, employer characteristics, cost quantities, severity classifications, and claim outcomes.

Workers' compensation claims are generally sorted into two categories: lost-time and medical-only. To focus on substantively complex cases, the analytical cohort was filtered to lost-time claims, which often involve indemnity payments due to extended work absence and more severe injuries. Medical-only claims typically apply to workers with minor injuries who return to work within 3 to 7 days and are therefore less likely to require legal assistance. Furthermore, the cohort was limited to five injury severity categories: Temporary Partial Disability, Temporary Total Disability, Permanent Partial Disability, Permanent Total Disability, and Death Benefits, excluding the medical-only severity type for consistency. Therefore, our findings apply to lost-time workers' compensation claims within the specified severity categories and should not be generalized to medical-only claims. After filtering, the final dataset contains 48,130 claims.

We define our target variable as the binary indicator of attorney involvement (`legal_or_attorney_ind`). In the filtered dataset, 26,659 claims (55.39%) did not involve legal representation, while 21,471 claims (44.61%) did, reflecting mild class imbalance. Although this distribution is nearly balanced, we specifically selected discrimination performance metrics that are robust against class imbalance.

To mimic realistic model deployment, we excluded features that are only available after claim escalation, such as legal identifiers (`attorney_name`, `legal_ind`, and the target) and cost-related variables (`indemnity_amt`, `medical_amt`, `dcc_amt`, `indweekkamt`, `medweekkamt`). We also removed post-outcome timing variables (`rtw_date`, `close_date`, `close_year`, `days_open_to_close`), and the claim status indicator (`claim_status`). The unique claim identifier (`uid_claim`) and administrative code (`waiting_period`) were also dropped to prevent memorization of claim-specific patterns. Lastly, `tenure_at_injury` was removed to reduce redundancy with the age variables.

The final dataset consists of 27 features, including 20 categorical attributes and 7 numerical variables. The categorical variables describe the severity, nature, affected body parts, and cause of the injury, occupation and industry classifications, marital status and gender of the employee, jurisdiction, and whether multiple workers were injured in the same event. The

numerical variables detail the employee’s age at injury, age at hire, weekly wage, employer’s payroll, credit score, number of dependents, and age of business. The retained predictors reflect data that would likely be available soon after a workers’ compensation claim is filed, preventing information leakage.

Missing values were handled natively by CatBoost, which incorporates missingness into its tree-splitting process to learn conditional relationships associated with incomplete samples [10]. This approach preserves the underlying data distribution and mitigates bias from alternative solutions, such as imputation or removal of missing entries. Finally, we partitioned 80% of the claims to the training set ($N = 38,504$) and 20% to the held-out test set ($N = 9,626$). These preprocessing procedures ensure our probabilistic modeling of attorney involvement aligns with realistic deployment conditions.

B. CatBoost and Hyperparameter Tuning

This study adopts a probabilistic modeling approach to estimate litigation risk and therefore requires a learning algorithm capable of producing reliable probability scores from structured, heterogeneous data. We selected the gradient boosting framework CatBoost for its strong performance with both categorical and numerical variables, as well as its native categorical handling [11], [12]. CatBoost uses ordered target statistics and permutation-based ordering for its categorical encoding. This approach avoids target leakage during training while preserving meaningful feature relationships. Because of these properties, CatBoost is a well-suited machine learning model for predicting the likelihood of attorney involvement and generating stable probability estimates.

The model hyperparameters were tuned to balance model capacity with generalization performance. The hyperparameter search considered the following predefined grid of configurations: the number of boosting iterations (500, 1000, 2000, and 3000), learning rate (0.03, 0.05, and 0.10), and tree depth (6 and 8). Selection was based on five-fold stratified cross-validation (CV) applied to the training set. Stratification preserves the outcome distribution of the overall training cohort within each fold. We used the area under the receiver operating characteristic curve (ROC-AUC), the area under the precision-recall curve (PR-AUC), and Brier score to evaluate the CV performance. The cross-validation results are not shown due to space limitations. The configuration settings with the highest mean cross-validated scores were 3000 iterations, a learning rate of 0.03, and a tree depth of 6. We selected these three hyperparameter values for our final CatBoost model. All other hyperparameters were left at their default settings.

C. Model Performance Metrics

After the hyperparameter selection and model training, we ran our CatBoost model on the held-out test set to assess its predictive performance. Because the goal of this study is to estimate escalation risk as a continuous probability, model evaluation focuses on both how well predictions are ordered and how accurately the predicted probabilities reflect actual outcomes. In other words, we measure discrimination and

calibration as complementary but distinct dimensions of model performance through four metrics: ROC-AUC, PR-AUC, Brier score, and Expected Calibration Error (ECE).

Discrimination performance was evaluated using ROC-AUC and PR-AUC. ROC-AUC represents the probability that a randomly selected claim involving attorney representation is assigned a higher predicted risk score than a randomly selected claim without representation [13]. Broadly speaking, ROC-AUC measures the agreement between predicted probabilities and observed outcomes across all possible classification thresholds, providing a threshold-independent assessment of ranking performance. PR-AUC complements this metric by centering on the trade-off between precision and recall, making it particularly informative in settings with class imbalance or when the positive class is the primary focus [14], [15].

Because our approach treats model outputs as direct estimates of escalation risk, we must evaluate calibration to ensure the probability scores reflect the true likelihood of attorney involvement [16], [17]. We selected the two metrics Brier score and ECE to assess calibration. The Brier score measures the mean squared difference between predicted probabilities and observed binary outcomes [16]. In Equation 1, we use $y_i \in \{0, 1\}$ to denote the true attorney involvement indicator for claim i , and $\hat{p}_i$ to denote the predicted probability of attorney involvement. The Brier score is defined as

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{p}_i)^2 \quad (1)$$

where N is the number of observations in the test set. Lower values indicate better probabilistic performance, with a value of zero corresponding to perfect prediction.

To summarize calibration deviation, ECE was computed using ten quantile-based bins [18]. In Equations 2 to 4, we use the variables B_m and $|B_m|$ to represent the set of observations in bin m and the number of observations within that bin, respectively. The accuracy and average predicted confidence for bin m are given by

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} y_i, \quad (2)$$

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i \quad (3)$$

The ECE is then defined as

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (4)$$

where $M = 10$ is the number of bins. Quantile-based binning ensures that each bin contains a similar number of observations, which improves stability in regions with sparse data. Lower ECE values indicate better alignment between predicted probabilities and observed outcomes.

D. Probabilistic Risk Ranking and Segmentation

The predicted probabilities are subsequently interpreted as continuous risk scores, providing the foundation for an ordered analysis of legal escalation risk. Claims are ranked in descending order according to these scores, establishing a relative hierarchy from highest to lowest estimated likelihood of attorney involvement without relying on a fixed decision threshold. This approach preserves variations in risk across claims and supports a more flexible analysis through segmenting claims into risk groups.

Building on this ordering, claims are partitioned into three risk regimes using a rank-based 20:60:20 segmentation. The bottom 20% of claims are classified as low risk, the top 20% as high risk, and the remaining 60% as medium risk. We devised the thresholds $q_{0.20}^{\text{rank}}$ and $q_{0.80}^{\text{rank}}$ to correspond to the predicted probabilities at the 20% and 80% rank positions, respectively. These boundaries are defined by the ordering of predicted risk rather than independently computed percentiles of the probability distribution. This design choice reflects a standard quantile-based partitioning strategy adopted for interpretability and balanced group sizes, instead of domain-specific empirical thresholds, resulting in risk regimes that are defined entirely by model-based estimates and can be applied without relying on observed outcomes.

E. Regime-Conditioned Explainability

Our explainability analysis investigates how the SHAP values behave across the low-, medium-, and high-risk regimes. For each claim i in regime r , the SHAP value ϕ_{ij} represents the contribution of feature j to the predicted probability of attorney involvement for that observation, relative to the model's baseline expectation. To summarize feature influence within each regime, we calculate the mean absolute SHAP value, which captures the average magnitude of a feature's effect regardless of direction:

$$\text{Magnitude}_j^{(r)} = \frac{1}{N_r} \sum_{i \in r} |\phi_{ij}| \quad (5)$$

where N_r denotes the number of claims in regime r . This measure captures how strongly a feature influences the model's predictions on average within a given risk group, independent of whether its effects are positive or negative.

To complement magnitude, we also examine the mean signed SHAP values within each regime to characterize directional behavior in terms of model contributions:

$$\text{Direction}_j^{(r)} = \frac{1}{N_r} \sum_{i \in r} \phi_{ij} \quad (6)$$

Positive values indicate that, on average, the feature increases model-predicted attorney involvement relative to the baseline within the regime, while negative values indicate a decreasing effect. Unlike the absolute measure, this quantity can cancel across the claims and therefore reflects the net directional influence rather than overall strength.

These two measures capture the strength and average direction of feature contributions to model predictions within each

regime. Conditioning SHAP analysis on risk regimes further reveals how model-attributed feature effects vary across different segments of the risk distribution. Features may exhibit changes in either magnitude, directional consistency, or both, indicating that their role in shaping model predictions is not uniform across the risk spectrum.

IV. RESULTS

A. Discrimination and Calibration Performance

CatBoost achieves a ROC-AUC of 0.7540 and a PR-AUC of 0.7148 on the test set, indicating strong performance in ranking claims by escalation risk. Probabilistic reliability is highly stable with a Brier score of 0.1988 and an ECE of 0.0114, suggesting minimal discrepancy between predicted probabilities and observed outcomes. These findings demonstrate that the model provides effective discrimination and produces well-calibrated probabilities. This validation step is important in our workflow, as the regime-conditioned explainability analysis builds on these predicted probabilities and their relative ranking rather than on classifications derived from fixed thresholds.

B. Risk Stratification

We next examine the full distribution of predicted probabilities produced by CatBoost across the 9,626 claims in the test set, as shown in Figure 1. The model outputs are interpreted as continuous risk scores and ordered from lowest to highest to form an escalation spectrum. Instead of imposing a fixed classification threshold, we partition claims by rank position into three regimes representing the bottom 20%, middle 60%, and top 20% of the predicted risk distribution.

The segmentation assigns 1,925 claims to the low-risk regime, 5,776 claims to the medium-risk regime, and 1,925 claims to the high-risk regime. The rank-based thresholds align with probability values of approximately 0.239 at the transition from low to medium risk and 0.646 at the transition from medium to high risk. Figure 1 displays the rank-based regime cutoffs at $q_{0.20}^{\text{rank}}$ and $q_{0.80}^{\text{rank}}$ as dashed lines in blue and red, respectively.

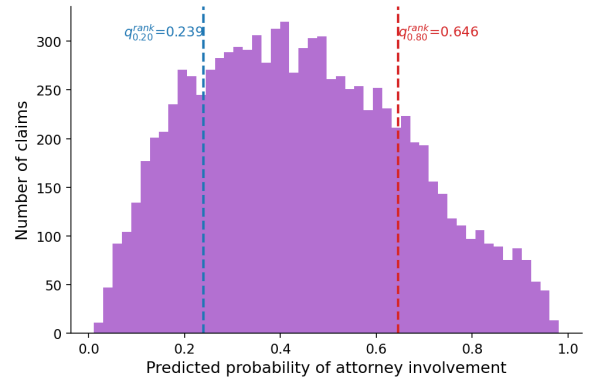


Fig. 1. Predicted risk score distribution with regime cutoffs at $q_{0.20}^{\text{rank}}$ and $q_{0.80}^{\text{rank}}$.

C. Regime-Conditioned Analysis

The ranking of influential features varies across the regimes in both magnitude and direction. Figure 2 contains three bar charts illustrating the top ten features for the low-, medium-, and high-risk levels with their corresponding mean absolute SHAP values. These plots highlight influential features in terms of magnitude. Figure 3 presents a multi-point dumbbell plot of mean signed SHAP values across the three risk levels, specifically including features with a maximum mean of at least $|0.05|$, and emphasizes average directional contributions. The features are in descending order based on their maximum mean SHAP values with pink, blue, and green markers representing low-, medium-, and high-risk groups, respectively.

Although the exact order varies across regimes, the features `injury_severity`, `age_at_hire`, `juris_st`, `age_at_injury`, and `bdyptr_desc` consistently rank as the top five based on their mean absolute SHAP values. In the low-risk group, the average directional contributions of the top ten features are negative, indicating that these features tend to reduce the predicted litigation risk relative to the baseline. In the medium-risk group, the directional effects are centered around zero, indicating that the positive and negative contributions largely offset each other within this regime. Although features such as `age_at_hire`, `age_at_injury`, and `bdyptr_desc` are highly ranked, their signed mean SHAP values are close to neutral, indicating limited directional influence on average. In the high-risk group, `injury_severity` has the highest mean absolute SHAP value overall and exhibits a nearly equal positive effect, demonstrating directional stability. Conversely, `age_at_injury` is ranked fourth for mean absolute SHAP values, but its signed contribution is nearly zero, indicating inconsistent directionality for high-risk claims. Similarly, the feature `nature_desc` notably appears in Figure 3, but not Figure 2, suggesting a lesser impact in magnitude yet reliable direction.

As illustrated in Figure 3, the predictors exhibit clear differences in average directional effects across the regimes. All twelve features in our analysis display negative mean SHAP contributions in the low-risk regime and positive contributions in the high-risk regime. This pattern reflects a shift in the conditional contribution of features to model predictions across segments of the risk distribution. In contrast, the medium-risk regime shows smaller and mixed signed contributions, suggesting weaker and less consistent directional influence. This analysis highlights regime-dependent variation in feature influence that is not captured by global importance measures.

V. DISCUSSION

Attorney involvement in workers' compensation claims is associated with a consistent set of influential factors. However, key predictors such as injury severity, jurisdiction, and employee demographics show regime-dependent magnitude and direction across the risk distribution. This structure is consistent with a staged pattern of escalation in model behavior. For low-risk claims, influential features tend to contribute to lower model-predicted risk. For example, injury severity

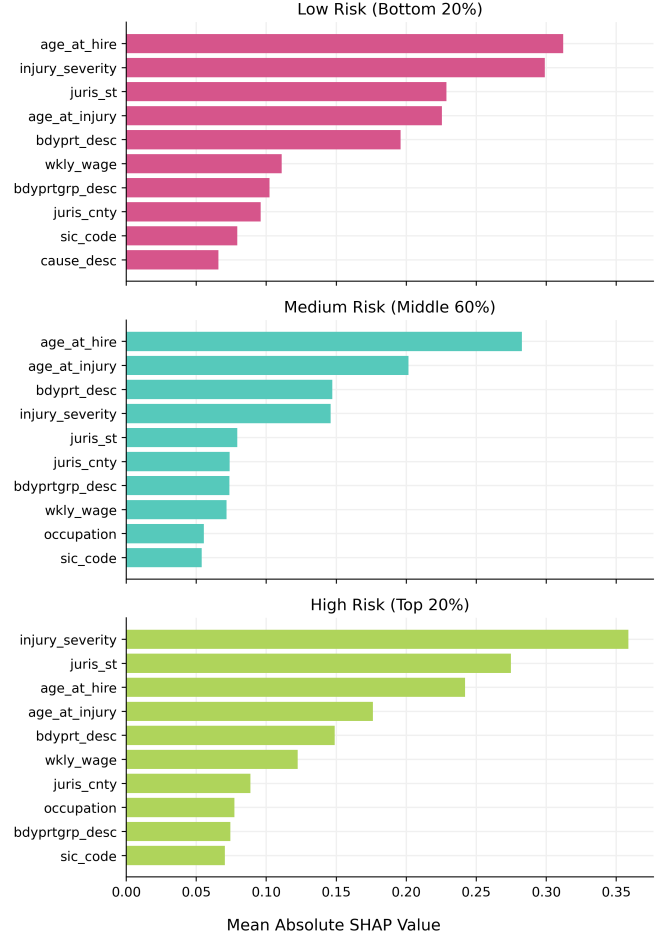


Fig. 2. Mean absolute SHAP values by risk regime.

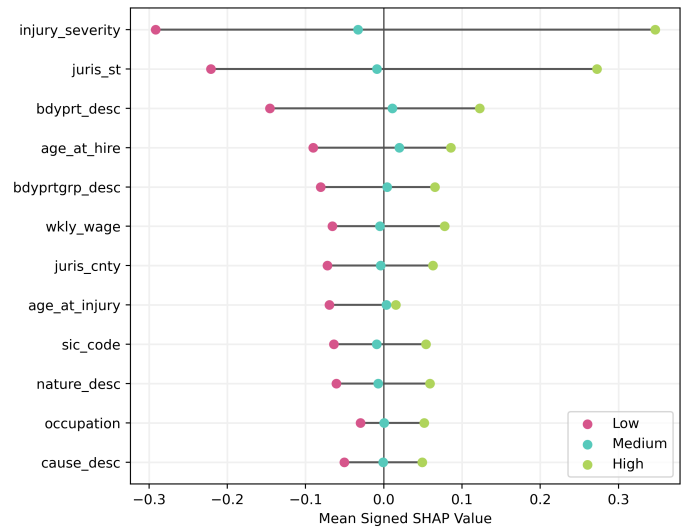


Fig. 3. Mean signed SHAP values across the risk regimes.

is the strongest negative contributor and second highest in magnitude for that regime. In medium-risk claims, feature effects reflect a reduced directional alignment in how predictors contribute to model outputs, as the mean signed SHAP values are concentrated near zero. In high-risk claims, the same predictors more reliably shift model predictions toward higher attorney involvement risk, with features such as injury severity becoming the dominant driver of escalation. These findings highlight the limitations of binary classification approaches that treat attorney involvement as a single binary outcome. Modeling risk as a continuous spectrum reveals meaningful heterogeneity in how predictors contribute to model predictions across different stages of litigation risk. For insurers and administrators, our results suggest that the low-, medium-, and high-risk regimes may require distinct management strategies, with particular attention to the medium-risk regime where predicted risk is most sensitive to contextual changes.

A few study limitations must be acknowledged. First, our analysis relies on a single workers' compensation dataset. Although this dataset is broader in jurisdictional coverage compared to many publicly available datasets, it does not fully capture the diversity of regulatory environments, which may limit generalizability given the predictive importance of jurisdiction. Second, due to space constraints, we limit our analysis to CatBoost and do not evaluate whether the feature importance patterns are consistent across alternative machine learning models. As SHAP values are defined relative to a model-specific baseline, our interpretations reflect the fitted CatBoost model and dataset. Lastly, although the model uses variables available early to mitigate information leakage, factors such as medical history, ongoing treatment plans, or legal availability are not captured in the current dataset and may significantly affect litigation outcomes.

VI. CONCLUSION

Our study presents a probabilistic approach for modeling attorney involvement in workers' compensation claims by treating escalation risk as a continuous outcome rather than a binary classification. Using a CatBoost classifier, we generate probability estimates and divide 9,626 claims into low-, medium-, and high-risk regimes based on their relative ranking. Through a regime-conditioned SHAP analysis, we examine how feature importance varies across the risk spectrum. Key predictors, such as age demographics, injury severity, and jurisdiction, are shown to influence predictions across the regimes, but their magnitude and direction are not uniform. The feature effects shift between the risk groups, presenting a pattern not captured by global importance measures.

These findings highlight how our risk stratification and SHAP analysis extends threshold-based classification to better capture dynamic feature contributions across the probability distribution. This work supports more targeted case management by identifying claims that may benefit from early intervention based on variables likely known at the time of model deployment. Our approach demonstrates how probabilistic modeling with regime-based explainability can provide deeper

insights for decision-making in complex workers' compensation claims. Future work could explore additional machine learning models and explore how feature contributions behave at the boundary cutoffs.

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